Mode-Path Matrix Quick Reference

Thermal Photonics & Diffusionics — Chapter 4 Companion

Thermal Photonics: A Unified Framework for Engineering Heat Flux
Book 3 of the Open-Source Engineering Optics Trilogy

`Version 3.0 | Standalone practitioner reference`

What To Do First

This is a **standalone desk reference** for the Mode-Path Matrix methodology. For the full derivation, worked examples, and proofs, see Chapter 4. For a first encounter, work through the WE-4.0 micro-example (referenced in §1.4) before tackling real systems.

1 The Thermal Network Paradigm

Any thermal system can be represented as a directed graph $G(V, E)$:

- **Vertices** V: thermal nodes (heat sources, sinks, junctions).
- **Edges** E: thermal conductances G [W/K] connecting nodes.

1.1 Conductance Formulae

$$G_{\text{cond}} = \frac{\kappa A}{L}, \qquad G_{\text{rad}} = 4\,\varepsilon\,\sigma_{\text{SB}}\,\bar{T}^3 A, \qquad G_{\text{conv}} = h\,A, \qquad G_{\text{contact}} = \frac{A}{R_c} \tag{1}$$

1.2 Thermal Kirchhoff's Laws

The following laws hold under three assumptions: (1) lumped-parameter approximation (Biot number $\text{Bi} = hL/\kappa < 0.1$ for each node), (2) linear constitutive relation $Q = G \cdot \Delta T$ (valid for conduction; for radiation, requires linearization—see DR-T.1b below), and (3) internal heat generation assigned to nodes, not edges.

Thermal Kirchhoff's Current Law (TKCL): At any node, conservation of energy requires

$$\sum_j G_{ij}(T_j - T_i) + Q_i = 0 \qquad \text{(steady state)} \tag{2}$$

For the transient case, the imbalance equals the stored energy rate: replace 0 with $C_i\, dT_i/dt$.

Thermal Kirchhoff's Voltage Law (TKVL): Temperature is a state function, so around any closed loop:

$$\sum_{\text{loop}} \Delta T_k = 0 \tag{3}$$

1.3 Matrix Formulation

Assembling TKCL for all nodes yields the system equation:

$$\mathbf{G} \cdot \mathbf{T} = \mathbf{Q} \qquad \text{(steady state)}, \qquad \mathbf{C}\,\frac{d\mathbf{T}}{dt} + \mathbf{G} \cdot \mathbf{T} = \mathbf{Q}(t) \qquad \text{(transient)} \tag{4}$$

where $\mathbf{G}$ is the conductance matrix, $\mathbf{T}$ the temperature vector, $\mathbf{Q}$ the source vector, and $\mathbf{C}$ the diagonal capacitance matrix.

1.4 Starter Example (WE-4.0 in Chapter 4)

Chapter 4 provides a 4-node micro-example (Die → TIM → Cu substrate → Heat sink) that explicitly assembles a 3×3 $\mathbf{G}$ matrix, solves $\mathbf{T} = \mathbf{G}^{-1}\mathbf{Q}$, identifies the critical path and bottleneck, and computes Γ for all edges. This is the recommended first exercise before applying the methodology to real systems. The solution requires only 10 lines of algebra and is verifiable with a pocket calculator.

2 Critical Path Analysis

2.1 Definitions

The **critical path** is the source-to-sink route with the highest total thermal resistance:

$$R_{\text{critical}} = \max_{\text{all paths } p} \left(\sum_{e \in p} R_e \right) = \max_{p} \left(\sum_{e \in p} G_e^{-1} \right) \tag{5}$$

2.2 Bottleneck Ratio

Within the critical path, the **bottleneck** is the element with the highest individual resistance. Quantify its dominance via:

$$\beta = \frac{R_{\text{bottleneck}}}{R_{\text{critical}}} \tag{6}$$

Table 1: Interpretation of the bottleneck ratio β.

β	**Interpretation**	**Design Action**
> 0.5	Single dominant bottleneck	Fix this element
0.2–0.5	Moderate bottleneck	Fix, then reassess
< 0.2	Distributed resistance	System-level optimization

3 The Mode-Path Matrix

3.1 Mode Dominance Ratio

For paths where conduction and radiation act in parallel:

$$\Gamma = \frac{G_{\text{rad}}}{G_{\text{cond}}} \tag{7}$$

Table 2: Mode classification via Γ.

Γ	**Dominant Mode**	**Design Toolkit**	**Book Reference**
< 0.1	**Conduction**	Diffusionics	Part II (Ch. 5–7)
0.1–10	**Mixed mode**	Multiphysics co-design	Ch. 9
> 10	**Radiation**	Thermal Photonics	Part III (Ch. 8–10)
$\to 0$ at cryo	**Conduction**	Cryogenic design	Ch. 14

At $\Gamma = 0.1$, the radiative contribution is less than 9%; at $\Gamma = 10$, the conductive contribution is less than 9%. In the mixed-mode band, neither mode can be neglected without introducing >9% error.

3.2 MPM Fill-Out Recipe

What To Do First

How to fill out a Mode-Path Matrix row by row:

1. **List every thermal path** from each source to each sink (DR-T.1). One row per path.
2. **Compute** $G_{\mathrm{cond}} = \kappa A/L$ for each path using material properties and geometry.
3. **Compute** $G_{\mathrm{rad}} = 4\,\varepsilon\,\sigma_{\mathrm{SB}}\,\bar{T}^{3}\,A$ using the linearized formula. Check $\Delta T/\bar{T} < 0.2$ (DR-T.1b); if violated, use exact $\varepsilon\,\sigma_{\mathrm{SB}}\,A\,(T_1^4 - T_2^4)$ and define an effective G_{rad}.
4. **Compute** $G_{\mathrm{conv}} = h\,A$ if convection is present (see §3.4). Record in the G_{conv} column.
5. **Compute** $\Gamma = G_{\mathrm{rad}}/G_{\mathrm{cond}}$ and classify using Table 2.
6. **Map each path** to the appropriate book chapter and simulation tool using the Tool Translation Table (Table 4).

3.3 Template Mode-Path Matrix

Table 3: Template Mode-Path Matrix for a generic thermal system.

Path	**From**	**To**	G_{cond} (W/K)	G_{rad} (W/K)	G_{conv} (W/K)	Γ	**Tool**
P1	Source	Spreader	50	0.5	—	0.01	Ch. 5–7
P2	Spreader	Ambient	2	8	5	4.0	Ch. 9
P3	Source	Lid	10	3	—	0.3	Ch. 9
P4	Lid	Ambient	1	5	20	5.0	BC only

3.4 Convection in the Mode-Path Matrix

In this book's two-mode taxonomy (phonon diffusion and photon emission), convection is *not* a third microscopic transport mode but rather an external boundary-layer conductance. The Mode-Path Matrix records G_{conv} for completeness but routes thermal design based on $\Gamma = G_{\mathrm{rad}}/G_{\mathrm{cond}}$ alone. Three rules govern its treatment:

1. **Boundary-only convection** (G_{conv} present only at system–ambient interface): Enter G_{conv} in the boundary-condition column. Design uses Γ for tool selection; convection sets the thermal resistance at the boundary.
2. **Internal forced flow** (e.g., coolant channels in a cold plate): Treat as a high-G conductive edge with $G_{\mathrm{eff}} = \dot{m}\,c_p$ or $h_{\mathrm{int}}\,A$. The path remains conduction-dominated ($\Gamma \ll 1$).
3. **Convection-dominated path** ($G_{\mathrm{conv}} \gg G_{\mathrm{cond}}$ and $G_{\mathrm{conv}} \gg G_{\mathrm{rad}}$): The path is not a meta-material design target. Mark as "BC only" in the Tool column. See Appendix D.

4 Tool Translation Table

Table 4: Tool Translation Table ("Rosetta Stone"): from Γ to design action. Given Γ for a path, read across the row to find your tool and your chapter.

Γ **Regime**	**Governing Physics**	**Simulation Tools**	**Book Ch.**	**One-Line Recipe**
< 0.1 (Cond.)	Fourier's law; $\mathbf{G}\cdot\mathbf{T} = \mathbf{Q}$	COMSOL, Ansys Icepak, FloTHERM, R-network	Part II (5–7)	Minimize R_{crit}; improve bottleneck $\kappa A/L$.
> 10 (Rad.)	Stefan–Boltzmann; radiosity $\mathbf{J} = f(\varepsilon, F_{ij}, T^4)$	Code V (non-seq), Zemax (non-seq), LightTools	Part III (8–10)	Modify ε or add spectral shield to control G_{rad}.
0.1–10 (Mixed)	Coupled cond.–rad.; iterative solve	COMSOL + radiation; Zemax + thermal; MPM-guided	Ch. 9	Solve coupled system; iterate with DR-T.6.
$\to 0$ at cryo	Kapitza cond.; $G_{\text{rad}} \propto T^3 \to 0$	Cryogenic FEA; custom MPM	Ch. 14	Invert objective: maximize parasitic-path R.

5 Three Limiting Cases

Most real systems fall into one of three simplified cases where the MPM reduces to a familiar tool:

1. **Photon-emission only** ($\Gamma \gg 1$ for all paths): The MPM reduces to the radiosity/view-factor matrix. Existing ray-tracing tools (Code V, Zemax, LightTools) solve the problem directly. Example: high-temperature furnace.
2. **Phonon-diffusion only** ($\Gamma \ll 1$ for all paths): The MPM reduces to the classical Fourier resistance network. Standard FEA tools (COMSOL, Icepak) solve it. Example: room-temperature IC package.
3. **Mixed photon + phonon** (Γ spans both regimes across different paths): The full MPM is required. No single tool covers all paths. Example: semiconductor fab with processes from 25 °C to 1100 °C.

The MPM is a *generalization* of these familiar methods, not a replacement. When all paths share one mode, the practitioner's existing tool suffices. The MPM becomes essential only in Case 3—and its first value is *telling you which case you are in.*

6 Conductance Matching

For two elements in series, define the **mismatch ratio**:

$$M = \frac{G_{\text{high}}}{G_{\text{low}}} = \frac{R_{\text{low}}}{R_{\text{high}}} \tag{8}$$

Table 5: Conductance mismatch severity.

M	**Severity**	**Consequence**
1–3	Well-matched	Efficient heat flow
3–10	Moderate	10–30% efficiency loss
10–100	Severe	Bottleneck-dominated
> 100	Extreme	One element irrelevant

The conductance matching principle interacts with the Mode-Path Matrix: if the bottleneck element is conduction-dominated ($\Gamma < 0.1$) but the adjacent element is radiation-dominated ($\Gamma > 10$), the appropriate design toolkits are *different*. The MPM prevents the error of applying conduction-based optimization to a radiation-limited interface.

7 System Design Workflow: The Six Steps

Multiphysics Integration

The Thermal Architecture Design Process:

1. **Define thermal requirements.** $T_{j,\max}$, power map $Q(\mathbf{r}, t)$, uniformity $\Delta T_{\max}$, response time τ_{proc}.
2. **Construct thermal network graph.** All nodes, all paths, initial conductance estimates (DR-T.1). Verify lumped-node validity (Bi < 0.1, Ch. 1).
3. **Analyze topology and find critical path.** Classify sub-networks (DR-T.2), identify highest-R route (DR-T.3), locate bottleneck via β.
4. **Build Mode-Path Matrix.** Calculate Γ for each path (DR-T.4). Check linearization validity (DR-T.1b). Use the Fill-Out Recipe (§3.2).
5. **Apply mode-appropriate design tools.** Critical path, conduction-dominated $\rightarrow$ Part II. Critical path, radiation-dominated $\rightarrow$ Part III. Mixed $\rightarrow$ Ch. 9. Check matching (DR-T.5).
6. **Verify and iterate.** Update network. Recalculate critical path—it may have shifted (DR-T.6). Repeat until all requirements are met.

8 Common MPM Failure Modes

Failure Mode:

Top Five Practitioner Mistakes

1. **Missing paths (violating DR-T.1).** Forgetting a radiative path (e.g., lens-to-housing gap) or a parasitic conduction path (e.g., cable bundles). Prevention: systematic node/edge enumeration before any calculation.
2. **Mode misidentification.** Assuming a path is conduction-only when $\Gamma = 2.5$ (mixed mode). Case study CS-6 in Ch. 12 shows this causes a 40% temperature prediction error. Prevention: always compute Γ numerically (DR-T.4).
3. **Linearization error (violating DR-T.1b).** Using $G_{\text{rad}} = 4\varepsilon\sigma_{\text{SB}}\bar{T}^3 A$ when $\Delta T/\bar{T} > 0.2$. The linearized formula under-predicts by >5%. Prevention: check $\Delta T/\bar{T}$ for every radiative edge.
4. **Optimizing the wrong element (violating DR-T.5).** Improving a via array ($G = 1000$ W/K) when the series TIM ($G = 1$ W/K) is the bottleneck ($M = 1000$). Prevention: compute M for adjacent pairs before design iteration.
5. **Not re-checking after improvement (violating DR-T.6).** After fixing the critical-path bottleneck, the critical path shifts to a different route. Case study CS-5 in Ch. 12. Prevention: always rebuild the MPM after every design change.

9 Navigation Routing Guide

After computing Γ for the critical-path bottleneck, use the following routing to find the appropriate design chapter:

- **If** $\Gamma < 0.1$ (conduction-dominated): Go to Part II. Ch. 5 for transformation thermotics and EMT; Ch. 6 for switchable/asymmetric devices; Ch. 7 for via arrays and planar architectures.
- **If** $0.1 < \Gamma < 10$ (mixed mode): Go to Ch. 9 (Multiphysics Coupling). You need coupled conduction–radiation design. Neither a pure FEA solver nor a pure ray tracer suffices alone.
- **If** $\Gamma > 10$ (radiation-dominated): Go to Part III. Ch. 8 for spectral engineering and selective emitters; Ch. 9 for coupled systems; Ch. 10 for thermophotovoltaics.
- **If** $\Gamma \to 0$ **at cryogenic** T: Go to Ch. 14 (Quantum-Thermal Frontier). The design objective *inverts*: maximize parasitic-path resistance rather than minimize cooling-path resistance.
- **If the bottleneck is convection (**G_{conv}**-limited)**: The path is not a metamaterial design target. See Appendix D for convection boundary conditions.

10 Robustness and Temporal Compatibility

Two final design rules close the methodology:

Design Rule:

DR-T.7 — Manufacturing Robustness Design the critical-path conductance with a 3σ margin:

$$G_{\text{design}} > G_{\text{required}} \times \left(1 + 3 \times \text{CV}\right) \tag{9}$$

where CV is the coefficient of variation (typically 0.05–0.15 for semiconductor packaging).

Design Rule:

DR-T.8 — Temporal Compatibility Verify that the thermal response time satisfies:

$$\tau_{\text{th}} < \tau_{\text{proc}} \tag{10}$$

If $\tau_{\text{th}} > \tau_{\text{proc}}$, the system cannot reach equilibrium during the process step, and transient analysis (Eq. 4, capacitance form) is mandatory. The critical path may differ between steady-state and transient regimes (cross-ref Chapter 2, DR-02.5).

11 Complete Design Rules Summary

Table 6: Complete set of topological design rules (DR-T series). Rules DR-T.1b and DR-T.4b added in v3.0 per reviewer consensus.

Rule	Key	Description	Eq.
DR-T.1	All paths	Construct complete thermal network graph identifying all heat flow paths. Verify lumped-node validity (Bi < 0.1).	—
DR-T.1b	$\Delta T/\bar{T}$	Verify $\Delta T/\bar{T} < 0.2$ before using linearized $G_{\text{rad}} = 4\varepsilon\sigma_{\text{SB}}\bar{T}^3 A$. Error exceeds 5% above this threshold; use exact Stefan–Boltzmann instead.	(1)
DR-T.2	Topology	Classify network topology (series, parallel, bridge, star, mesh, tree, hierarchical) before selecting components.	—
DR-T.3	Highest R	Identify and prioritize critical path; locate bottleneck via β.	(5)
DR-T.4	Γ	Build Mode-Path Matrix for *all* paths; use Γ for tool selection. **Most important DR in this book.**	(7)
DR-T.4b	All paths, not just dominant	Even when the critical path is conduction-dominated, compute Γ for *every* path. Radiative parasitic paths may cause transient asymmetries exceeding optical tolerances.	—
DR-T.5	$M < 10$	Maintain conductance matching between adjacent series elements. If $M > 10$, improving the high-G element yields $< 10\%$ benefit.	(8)
DR-T.6	Iterate	After critical-path improvement, recalculate network. The critical path may shift.	—
DR-T.7	3σ margin	Design with manufacturing robustness.	(9)
DR-T.8	$\tau_{\text{th}} < \tau_{\text{proc}}$	Verify temporal compatibility.	(10)

12 Cross-Reference Guide (v3.0 Chapter Map)

Table 7: Mode-Path Matrix applications across the book (v3.0 chapter numbering).

Chapter	**Mode-Path Matrix Application**
Ch. 1 (Dual Nature of Heat)	Fourier–Snell bridge: thermal field $\rightarrow$ GRIN profile $\rightarrow$ wavefront error. Motivates why system-level MPM is needed.
Ch. 2 (Timescales of Control)	Temporal regime classification; dynamic $\Gamma(t)$ during transients. DR-T.8 temporal compatibility.
Ch. 3 (Fluctuations & Near-Field)	Near-field G_{rad} enhancement modifies Γ at sub-λ_{th} gaps in 3D packaging.
Ch. 5 (Transformation & EMT)	Transformation devices modify local G; EMT simplifies network by replacing microstructure with effective conductance.
Ch. 6 (Switchable Devices)	Doublets reconfigure network topology; diodes add nonlinear (asymmetric) conductance.
Ch. 7 (Via Arrays)	Via arrays as high-G network edges; clustered vias as parallel conductances to hotspot nodes.
Ch. 8 (Spectral Engineering)	Linearized $G_{\text{rad}} = 4\varepsilon\sigma_{\text{SB}}T^3A$ as network element; spectral selectivity modifies ε.
Ch. 9 (Multiphysics)	Mixed-mode paths ($0.1 < \Gamma < 10$); coupled conduction–radiation network. Tool Translation Table maps Γ to Code V / COMSOL.
Ch. 10 (Thermophotovoltaics)	Radiation-dominated energy conversion paths; spectral control of G_{rad} for power generation.
Ch. 11 (Semiconductor Control)	Network model: plasma $\rightarrow$ wafer $\rightarrow$ chuck $\rightarrow$ coolant; chuck design via MPM.
Ch. 12 (Failure Analysis)	Critical path shift (CS-5), mode misidentification (CS-6); four-axis I/M/F/T failure framework.
Ch. 13 (Manufacturing)	Fabrication tolerance impact on G distributions; connects to DR-T.7 (3σ margin).
Ch. 14 (Quantum-Thermal)	Dual MPM (cooling vs. parasitic); $\Gamma \rightarrow 0$ at mK temperatures except stray IR through viewports.
Ch. 15 (Active Future)	Adaptive thermal networks; ML-optimized topology; differentiable MPM.

Annotated References

Table 8: Key references and their connection to the Mode-Path Matrix framework.

Reference	**MPM Connection**
Kirchhoff (1847)	Foundation: electrical network laws that thermal networks inherit (TKCL, TKVL).
Incropera et al., *Fundamentals of Heat and Mass Transfer*	Standard thermal resistance network methodology; the "phonon-only" limiting case of the MPM.
Modest, *Radiative Heat Transfer*	Radiosity and view-factor methods; the "photon-only" limiting case. Provides G_{rad} formulae.
Howell, Menguc, Siegel, *Thermal Radiation Heat Transfer*	Enclosure analysis, configuration factors; direct input to the radiation columns of the MPM template.
Fan & Huang (2008), *Shaped graded materials*	Transformation thermotics: the design tool for conduction-dominated critical paths (Part II).
Xu & Huang, *Transformation Thermotics and Extended Theories* (2023)	Comprehensive diffusionics framework; theoretical foundation for Part II design rules.
Kelley & Walker (1959), CPM	Critical path method from project management; adapted here for thermal network bottleneck analysis.
Cauer (1926), network synthesis	RC network synthesis theory; mathematical basis for the $\mathbf{G}\cdot\mathbf{T}=\mathbf{Q}$ formulation and eigenvalue decomposition of transient networks.

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Cross-references: Eikonal Bridge (ray–wave duality in propagation);
Quantum Field Imaging (field correlations in detection)